Practice Problems Chapter-8

1. Sequence : 1 0 1 1
   D_0 D_1 D_2 D_3

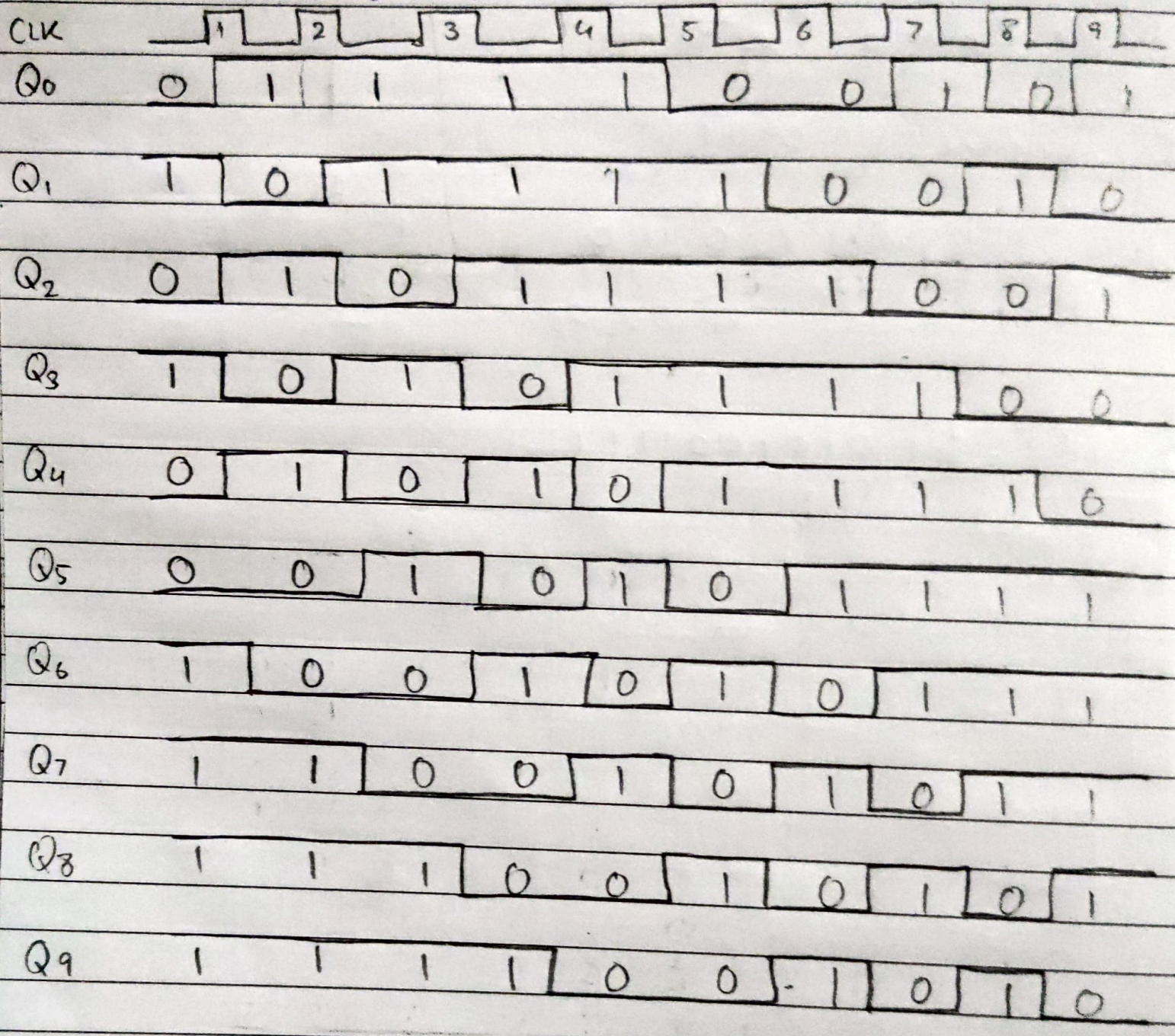
After 3 clock pulses : 0 1 1 0
↑ ↑ ↑ ↑
Q_0 Q_1 Q_2 Q_3

	Q_0	Q_1	Q_2	Q_3
Init:	0	0	0	0
CLK1:	1	0	0	0
CLK2:	1	1	0	0
CLK3:	0	1	1	0
CLK4:	1	0	1	1

Date: M T W T F S S

Q_0 ↓ Q_9 ↑

2. Initial State: 0 1 0 1 0 0 1 1 1 1



Date:
M T W T F S S

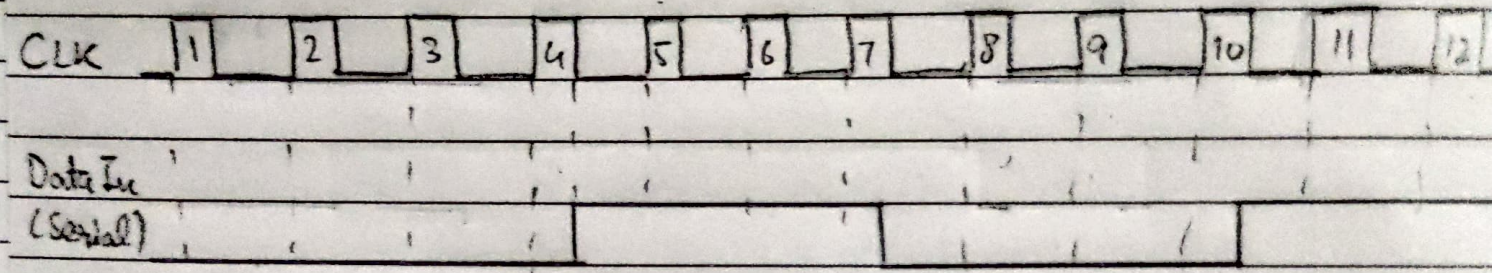
3. 8-bit Johnson counter

2(8) = 16 states

	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7
0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0
2	1	1	0	0	0	0	0	0
3	1	1	1	0	0	0	0	0
4	1	1	1	1	0	0	0	0
5	1	1	1	1	1	0	0	0
6	1	1	1	1	1	1	0	0
7	1	1	1	1	1	1	1	0
8	1	1	1	1	1	1	1	1
9	0	1	1	1	1	1	1	1
10	0	0	1	1	1	1	1	1
11	0	0	0	1	1	1	1	1
12	0	0	0	0	1	1	1	1
13	0	0	0	0	0	1	1	1
14	0	0	0	0	0	0	1	1
15	0	0	0	0	0	0	0	1
16	0	0	0	0	0	0	0	0

Q7:

Initial State: 10100111000



Initial State Column

Date:
M T W T F S S

Initial State: 1 0 1 0 0 1 1 1 1 0 0 0

After:

CLK 1: 0 1 0 1 0 0 1 1 1 1 0 0

CLK 2: 0 0 1 0 1 0 0 1 1 1 1 0

CLK 3: 0 0 0 1 0 1 0 0 1 1 1 1

CLK 4: 0 0 0 0 1 0 1 0 0 1 1 1

CLK 5: 1 0 0 0 0 1 0 1 0 0 1 1

" 6: 1 1 0 0 0 0 1 0 1 0 0 1

" 7: 1 1 1 0 0 0 0 1 0 1 0 0

" 8: 0 1 1 1 0 0 0 0 1 0 1 0

" 9: 0 0 1 1 1 0 0 0 0 1 0 1

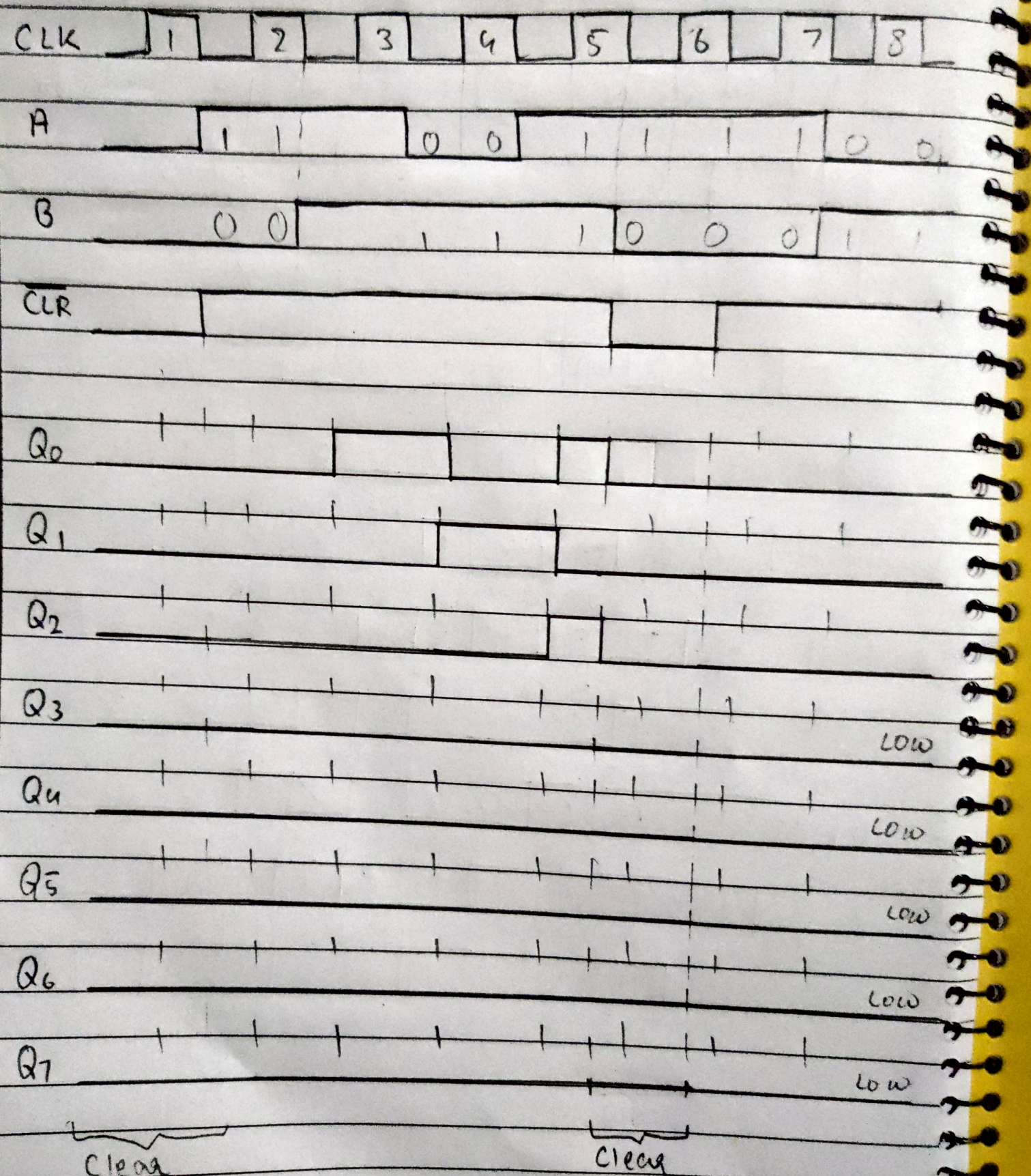
" 10: 0 0 0 1 1 1 0 0 0 0 1 0

" 11: 1 0 0 0 1 1 1 0 0 0 0 1

" 12: 1 1 0 0 0 1 1 1 0 0 0 0

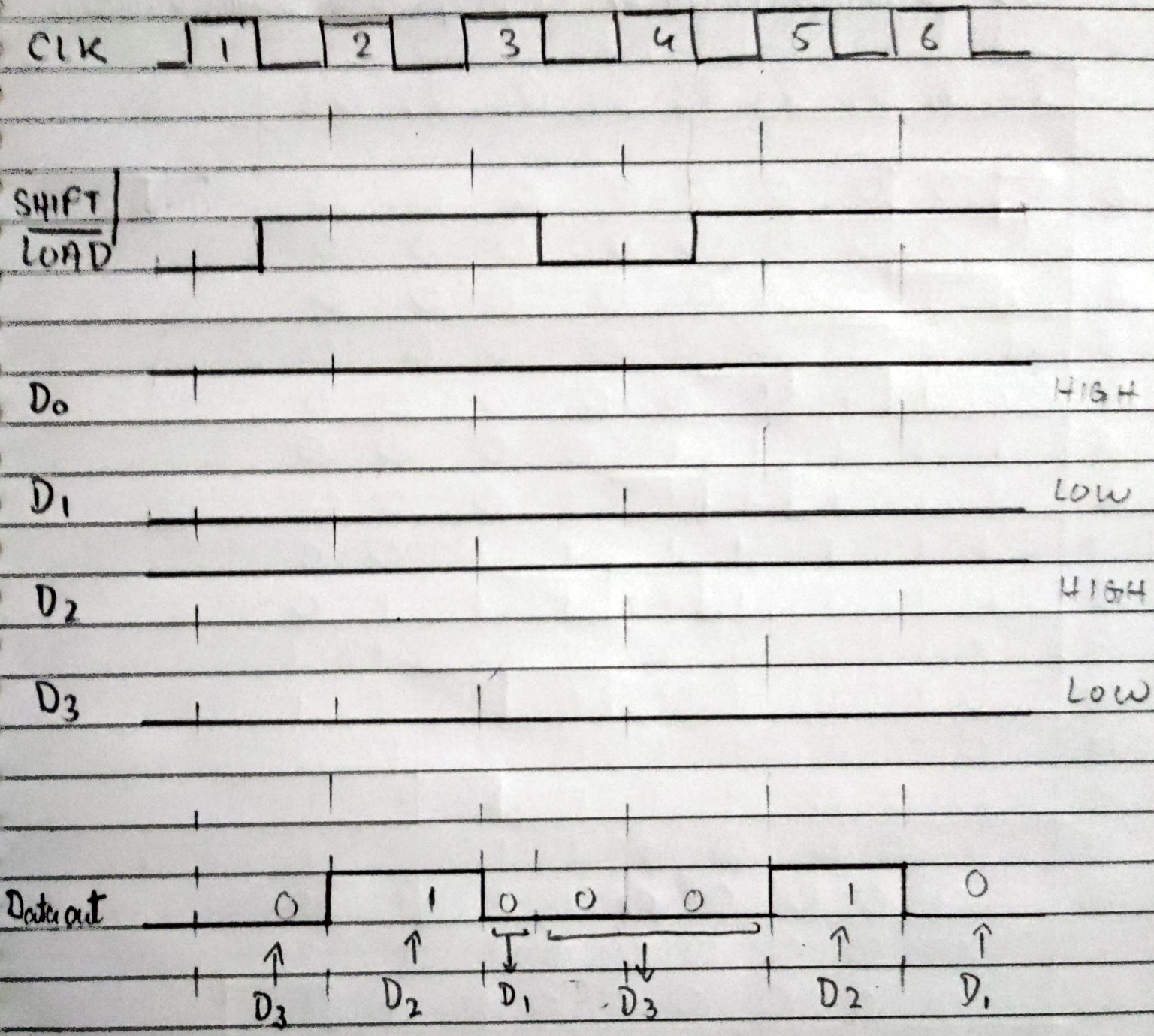
Date:
M T W T F S S

Q13.



Date:
M T W T F S S

Q14.



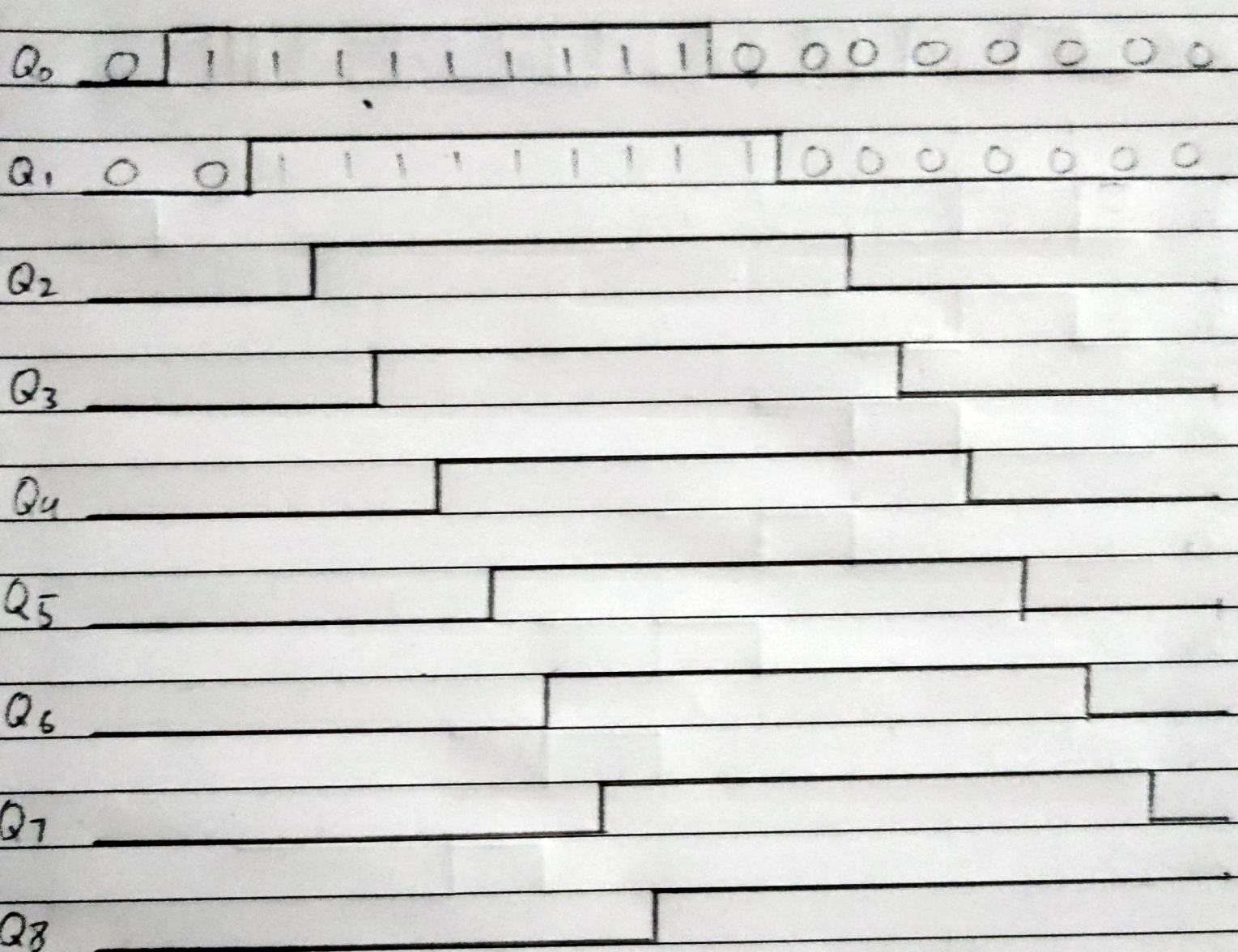
Q26;

Mod - 18 Johnson counter ⇒ 9 Flip Flops to be used (18 states).

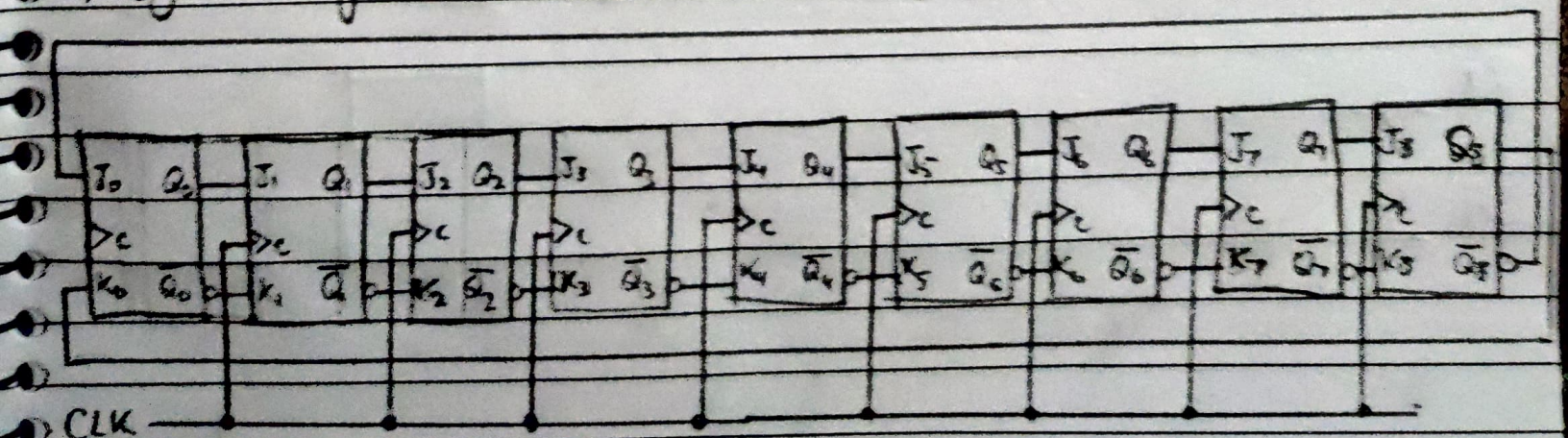
	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0
2	1	1	0	0	0	0	0	0	0
3	1	1	1	0	0	0	0	0	0
4	1	1	1	1	0	0	0	0	0
5	1	1	1	1	1	0	0	0	0
6	1	1	1	1	1	1	0	0	0
7	1	1	1	1	1	1	1	0	0
8	1	1	1	1	1	1	1	1	0
9	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
11	0	0	1	1	1	1	1	1	1
12	0	0	0	1	1	1	1	1	1
13	0	0	0	0	1	1	1	1	1
14	0	0	0	0	0	1	1	1	1
15	0	0	0	0	0	0	1	1	1
16	0	0	0	0	0	0	0	1	1
17	0	0	0	0	0	0	0	0	1
18	0	0	0	0	0	0	0	0	0

Date:
M T W T F S S

TIMING DIAGRAM:



→ Logic Diagram:



Date:
M T W T F S S

Q.27:

